

μOS++ III harness and tests

Building the test workspace in /home/dan/TMP, step by step

Kernel: micro-os-plus-iii

Ports: aarch32, aarch64, cortexm, posix-arch

/home/dan/TMP — harness + tests, step by step

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/home/dan/TMP — harness + tests, step by step

Reproducible log of building the μ OS++ test harness in `/home/dan/TMP`, with the tests of the six repositories wired into the tests paradigm, named `qemu` (runs under QEMU) or `hwd` (runs on hardware), and runnable from the VS Code xPack plugin.

Everything below uses `$HOME` and `$PWD` only — no hard-coded user — so it reproduces anywhere on this machine.

what	path
bare mirrors (clone source)	<code>\$HOME/Downloads/GIT/micro-os-plus-iii-*.git</code>
this workspace	<code>\$HOME/TMP</code>
node / npm / xpm	<code>node v25.8.2</code> , <code>npm 11.11.1</code> , <code>xpm 0.23.2</code>

Step 0 — the workspace

```
mkdir -p "$HOME/TMP"
cd "$HOME/TMP"
```

Step 1 — clone the six repositories

The harness lives in `micro-os-plus-iii-smp.git/tests/`, and it finds the ports as siblings named `micro-os-plus-iii-aarch32.git` and `...-aarch64.git` (see `tests/cmake/tests-main.cmake`), so the clones keep the `.git` suffix.

```
cd "$HOME/TMP"
for r in aarch32 aarch64 cortexm devices posix-arch smp; do
  git clone "$HOME/Downloads/GIT/micro-os-plus-iii-$r.git" \
    "$HOME/TMP/micro-os-plus-iii-$r.git"
done
```

Step 2 — sibling naming (no symlinks)

Two naming conventions meet here:

- the **harness** looks for the ports **with** the suffix (`../../micro-os-plus-iii-aarch32.git`);
- a **port** looks for its dependencies by the plain name (`../micro-os-plus-iii-smp`, `../micro-os-plus-iii-devices`).

The ports now accept **either**: if the plain-name sibling is absent and the `*.git` one is present, they use it — a `.git` fallback in each port's `CMakeLists.txt` (`UOS_SMP_DIR` / `UOS_DEVICES_DIR`) and in the port test scripts' `SMP_DIR` . So the workspace keeps **only** the `.git` clones and needs no suffix-less symlinks:

```
cd "$HOME/TMP"
ls -d micro-os-plus-iii-*.git
```

The harness itself already used the `.git` paths, so nothing else changes.

Step 3 — the harness tools

```
cd "$HOME/TMP/micro-os-plus-iii-smp.git/tests"
npm install      # node_modules
xpm install      # xpacks/ symlinks (cmake, ninja, build-helper, validator, chan-fatfs)
```

Step 4 — one configuration, end to end

Every configuration is driven by four `xpm run` actions, in this order: `install`, `prepare`, `build`, `test` . They are the `cmake-actions` in `tests/package.json`, and its `properties` hold the exact commands:

action	runs	which is
install	xpm install --config <config>	the pinned toolchain into build/<config>/xpacks/.bin
prepare	cmake -S . -B build/<config> -G Ninja -D CMAKE_BUILD_TYPE=... -D PLATFORM_NAME=... --log-level=VERBOSE -D CMAKE_TOOLCHAIN_FILE=xpacks/@micro-os-plus/build-helper/cmake/toolchains/<file>	commandCMakePrepareWithToolchain
build	the same cmake ... without the toolchain file, then cmake --build build/<config>	commandCMakeReconfigure + commandCMakeBuild
test	cd build/<config> && ctest -V (each config narrows this to ctest -V -LE hwd)	commandCMakePerformTests
clean	cmake --build build/<config> --target clean	commandCMakeClean

Three rules follow, and each has bitten:

- **install before prepare**. `prepare` names a toolchain file that lives *inside* the build folder (`build/<config>/xpacks/...`); before `install` it is not there and the configure fails.
- **prepare before build**. Only `prepare` passes `-D CMAKE_TOOLCHAIN_FILE=...`. The `build` action's reconfigure does **not**, so a build folder that was never prepared configures with the host compiler — or the wrong cross compiler. (`prepare` is idempotent; the `build` reconfigure keeps the cached toolchain, so once prepared, builds are fine.)
- **test does not build**. `xpm run test` on a folder that was only configured reports every case as failed, because the executables were never linked:

```
no *-host executables in .../platform-bin/port-tests
```

Run `build` first. The per-suite actions are the exception — each is self-contained (`commandCMakePrepareWithToolchain && commandCMakeBuild && ctest -V -R <case>`), so `xpm run test-rtos-apis-host --config native-cmake-gcc-debug` builds and runs that one case on its own. That is why the VS Code plugin's per-test actions work on a cold tree while the generic `test` does not.

The pinned toolchain is installed **inside the build folder** (`build/<config>/xpacks/.bin`), so `build/<config>/` must not be deleted. If the cache is ever poisoned (e.g. by a configure that ran before `install`), delete only the cache, not the whole folder:

```
cd "$HOME/TMP/micro-os-plus-iii-smp.git/tests"
xpm run install --config aarch32-rpi-zero-2w-cmake-gcc-debug
xpm run prepare --config aarch32-rpi-zero-2w-cmake-gcc-debug
xpm run build --config aarch32-rpi-zero-2w-cmake-gcc-debug
xpm run test --config aarch32-rpi-zero-2w-cmake-gcc-debug
```

Debug and release are separate build folders

`-release` is not a mode of the debug folder. Every configuration is its own `build/<config>/` — there are 62 of them (31 debug + 31 release) — and a release config inherits the debug config's *properties* but **not** its build folder, so it needs its own `install`. Skip that and `prepare` fails as soon as the dependencies are added:

```
CMake Error at xpacks/@micro-os-plus/build-helper/cmake/micro-os-plus-build-helper.cmake:112
(message):
  Missing ../build/native-cmake-clang17-release/xpacks/@xpack-3rd-party/libucontext/CMakeLists.txt
```

`native-dependencies` brings in `libucontext` (and `posix-arch`), so that folder has to exist before the configure; it is empty when `install` was never run. The clang configs also pin their own compiler — `native-cmake-clang17-release` wants `clang 17.0.6-3.1`, not whatever `clang` is on `PATH`.

All the release configurations

The emulated/host set, release, four steps each:

```
cd "$HOME/TMP/micro-os-plus-iii-smp.git/tests"
for C in \
  native-cmake-gcc-release \
  qemu-cortex-m0-cmake-gcc-release qemu-cortex-m3-cmake-gcc-release \
  qemu-cortex-m4f-cmake-gcc-release qemu-cortex-m7f-cmake-gcc-release \
  cortexm-pico2-cmake-gcc-release \
  cortexm-pico2-rp2350b-psram-cmake-gcc-release \
  aarch32-rpi-zero-2w-cmake-gcc-release aarch64-rpi-zero-2w-cmake-gcc-release ; do
  xpm run install --config "$C" && xpm run prepare --config "$C" && \
  xpm run build --config "$C" && xpm run test --config "$C"
done
```

The hardware set, release, build only — their cases are all `hwd`:

```

for C in \
  cortexm-pico2-pizero-cmake-gcc-release \
  cortexm-nucleof411-cmake-gcc-release cortexm-weactf411-cmake-gcc-release \
  cortexm-weactf412-cmake-gcc-release \
  nucleo-f411re-cmake-gcc-release nucleo-f767zi-cmake-gcc-release \
  nucleo-h743zi-cmake-gcc-release raspberrypi-pico-cmake-gcc-release ; do
  xpm run install --config "$C" && xpm run prepare --config "$C" && \
  xpm run build --config "$C"
done

```

The host toolchain variants are the same recipe with a different toolchain file: `native-cmake-{gcc,gcc11,gcc12,gcc13,gcc14,clang,clang13,clang14,clang15,clang16,clang17,clang18,clang19,sys}-release`. `aarch32-luckfox-lyra-cmake-gcc-release` builds its all-`hwd` set.

Step 5 — put all the port's tests into the paradigm

The port owns its tests (`micro-os-plus-iii-aarch32.git/test/rpi-zero-2w/<app>/`) and its own builder (`test/CMakeLists.txt`), which compiles each test twice — `<app>-qemu` and `<app>-hwd` . The port adds that builder only when it is the top-level project (`PROJECT_IS_TOP_LEVEL`), so the harness normally re-lists a few tests by hand.

Instead, the platform now **reuses the port's builder** and registers what it produced, so every test the port has appears, and a test added to the port appears with no edit. In `tests/platforms/aarch32-rpi-zero-2w/CMakeLists.txt` :

```

include ("cmake/platform-library.cmake")

# The port publishes most board facts as CACHE INTERNAL, but not
# UOS_BOARD_DEVICES; read the board declaration so the builder sees it.
include ("${UOS_AARCH32_DIR}/test/boards/${BOARD}/board.cmake")

# Build every port test (qemu + hwd) by reusing the port's builder.
add_subdirectory ("${UOS_AARCH32_DIR}/test" "port-tests")

set (_port_bin "${CMAKE_CURRENT_BINARY_DIR}/port-tests")
set (_shim_img "${_port_bin}/shim8.img")      # the builder's QEMU boot shim
set (_runner   "${CMAKE_SOURCE_DIR}/../test_smpl")

# One CTest case per (test, variant): <platform>-<app>-qemu / -hwd.
file (GLOB _test_dirs LIST_DIRECTORIES true "${_port_test_dir}/*")
foreach (_dir IN LISTS _test_dirs)
    ... # skip include/ and src/, keep the directories that hold sources
    add_test (NAME "${PLATFORM_NAME}-${_app}-qemu"
        COMMAND bash "${_runner}/run-qemu.sh" "${_port_bin}" "${_qemu}" -M raspi3b -smp 4)
    set_tests_properties ("${PLATFORM_NAME}-${_app}-qemu" PROPERTIES
        ENVIRONMENT "UOS_QEMU_SHIM=${_shim_img};UOS_QEMU_LOAD_ADDR=0x10000;UOS_QEMU_ONLY=${_app}"
        LABELS qemu)
    add_test (NAME "${PLATFORM_NAME}-${_app}-hwd"
        COMMAND bash "${_runner}/run-hw.sh" "${_port_bin}" "${_app}" 600)
    set_tests_properties ("${PLATFORM_NAME}-${_app}-hwd" PROPERTIES
        ENVIRONMENT "BUILD=${_port_bin}" LABELS hwd)
endforeach ()

```

The run side is the **shared runner** (micro-os-plus-iii-smp.git/test_smpl/) that the port's own `qemu.sh` / `hw.sh` call — `run-qemu.sh` greps the `RESULT:` line, `run-hw.sh` drives OpenOCD and takes one test (one per power cycle).

Step 6 — add the harness suites to the platform

A harness suite supplies no `main()` / hooks, so it links `micro-os-plus::platform-support`. It ends through the platform's strong `_Exit()` (SYS_EXIT), so QEMU returns an exit code and there is no `RESULT` line to grep — the direct QEMU command, not `run-qemu.sh`.

```
function (add_harness_suite _suite _name)
  add_executable (${_name})
  target_link_libraries (${_name} PRIVATE micro-os-plus::common-options
    test::${_suite} micro-os-plus::iii micro-os-plus::platform
    micro-os-plus::platform-support ${T_LIBRARIES})
  set_target_properties (${_name} PROPERTIES OUTPUT_NAME "${_name}" SUFFIX "")
  add_test (NAME "${PLATFORM_NAME}-${_name}-qemu"
    COMMAND "${_qemu}" --machine raspi3b -smp 4 --nographic --serial none
      --semihosting-config enable=on,target=native
      --kernel "${_shim_img}"
      --device "loader,file=${<TARGET_FILE:${_name}>.bin,addr=0x10000")
endfunction ()

add_harness_suite (mutex-stress mutex-stress-test)
```

Step 7 — verify

```
cd "$HOME/TMP/micro-os-plus-iii-smp.git/tests"
B=build/aarch32-rpi-zero-2w-cmake-gcc-debug
PATH="$PWD/$B/xpacks/.bin:$HOME/.local/xPacks/@xpack-dev-tools/qemu-arm/9.2.4-1.1/.content/bin:
$PATH" \
  ctest --test-dir "$B" -N                                # list
PATH=... ctest --test-dir "$B" -R "smp_test0-qemu|mutex-stress" -V
```

Result: **26 CTest cases** — the port's tests as `qemu / hwd` (14 `qemu` + 12 `hwd`), plus the `mutex-stress` and `rtos-apis` suites. Verified passing: `smp_test0-qemu`, `smp_test1-qemu`, `smp_test3-qemu`, `mutex-stress-test-qemu`, `rtos-apis-test-qemu`.

Status, and what is still to do

Done

- the workspace, the six clones (`.git` only — no suffix-less symlinks; the ports accept either name);
- the harness tools, and the configurations end to end;
- `platforms/aarch32-rpi-zero-2w` — **all** the port's tests as `<platform>-<app>-qemu / -hwd`, plus `mutex-stress` and `rtos-apis` (26 cases: 14 `qemu` + 12 `hwd`). Verified passing: `smp_test0/1/3-qemu`, `mutex-stress-test-qemu`, `rtos-apis-test-qemu`.
- `platforms/aarch32-luckfox-lyra` — the port's 19 tests + `mutex-stress`, `-hwd` **only** (the board is hardware-only); 20 CTest cases, never run by a plain `ctest`.

- `platforms/aarch64-rpi-zero-2w` — the port's tests × `qemu` / `hwd`, plus `mutex-stress` and `rtos-apis` (26 cases: 14 `qemu` + 12 `hwd`); verified passing: `smp_test0-qemu`, `mutex-stress-test-qemu`, `rtos-apis-test-qemu`. The `_Exit()` link issue is **fixed**: the port's own `src/handlers.cpp` already defines a strong semihosting `_Exit` / `_exit`, so the platform-support's copy was a duplicate definition; it was removed and the suite enabled.
- `platforms/native` — the **POSIX-arch** port on the **host compiler** (`micro-os-plus-iii-posix-arch.git`), selected by the new `^native` branch in `tests-main.cmake`; 12 host tests (the port's 11 + `cmsis-os-validator`), verified passing: `smp_test0-host`, `cmsis-os-validator-host`.

The remaining harness suites — now wired and passing.

- `rtos-apis` — **now wired and passing on both AArch32 and AArch64** (`...-rtos-apis-test-qemu`, ~7 s each). Three things were wrong:
 - **`lifo::do_allocate` bug (kernel)**. It set `chunk = free_list_` and, when that head chunk was smaller than the request (`rem < 0`), never cleared `chunk` — so the loop broke and handed the too-small chunk to `internal_align_`, which asserts. `first_fit_top` advances `chunk = chunk->next` and is correct; `lifo` now sets `chunk = nullptr` on the too-small path, so it falls through to the out-of-memory handler (which may return to coalesce) or returns `nullptr`.
 - **Undersized RTOS dynamic memory**. `sources/rtos-apis/.../os-app-config.h` set `OS_INTEGER_RTOS_DYNAMIC_MEMORY_SIZE_BYTES` to 14 KiB, but the default resource is that system heap, and FatFs asks for up to `MAX_MALLOC` (0x8000) — with the out-of-memory hook fatal (no exceptions), the test aborted. Raised to 512 KiB.
 - **AArch64 newlib signature clash**. `aarch64-none-elf` declares `read` / `write` with `_READ_WRITE_RETURN_TYPE` (`int`), the kernel header used `ssize_t`. `c-syscalls-aliases-standard.h` now uses `_READ_WRITE_RETURN_TYPE` when the system header provides it. The AArch64 platform also defines `OS_USE_SEMIHOSTING_SYSCALLS` so the tests skip their raw-POSIX-C-API sub-tests (this port does not carry that layer).
- `cmsis-os-validator`, `blinky`, `instrumentation` — **now build**. The `nucleo-f411re` platform (the upstream Cortex-M one, already in the tree) wires all three, plus `rtos-apis` and `mutex-stress`. They did not build before only because the SEGGER xpacks were missing:
 - `xpm install --config nucleo-f411re-cmake-gcc-debug` now installs `@xpack-segger/rtt@8.56.1-2` and `@xpack-segger/system-view@3.60.5-2` from `github:xpack-3rd-party/...`; the platform then builds `rtos-apis-test.elf`, `rtos-apis-instrumentation-test.elf`, `mutex-stress-test.elf`, `cmsis-os-validator-test.elf`, `blinky-test.elf` and `instrumentation-test.elf`.
 - `cmsis-os-validator` needs the NVIC API, which the `nucleo-f411re` platform-support provides (CMSIS core); `blinky` gets its `blink-led.h` from the platform (PA5), and `instrumentation` links `segger::rtt` + `segger::system-view`.
- All four of its `add_test`s drive the board through OpenOCD, so they are now labelled `hwd` (a directory-level `set_tests_properties`); the generic `test` action skips them, and `test-nucleo-f411re-cmake` runs them on a bench (`ctest -V`).
- **Native `cmsis-os-validator` — now passing 60/60** (`native-cmsis-os-validator-host`, ~8 s). The `native` platform builds and runs the validator (the xpack's `main.c` shims `NVIC_*` with `SIGUSR1`). Two real port/kernel bugs had to be fixed, plus three build fixes:

- **ISR context.** The POSIX-arch port's `in_handler_mode()` only looked at its own `_in_isr[]`, which its tick/IPI handlers set; the validator's `SIGUSR1` handler instead bumps `signal_nesting`. `TC_ThreadInterrupts` therefore saw thread mode and got the wrong "called from ISR" return codes. The port now defines `signal_nesting` and `in_handler_mode()` returns `_in_isr[cpu] || signal_nesting != 0` (the ARM ports have no banked mode to ask either).
- **Deferred wake-up.** `message_queue::send()` (used by `osMessagePut` with `osWaitForever`) and its `try_send` / `receive` / `try_receive` / `timed_receive` siblings called `internal_try_*` — which resumes a waiter — *inside* their critical section. The port cannot switch while the kernel lock is held, so it deferred to the next 1 ms tick, and `TC_MsgQWait`'s immediate assert failed. Each now reschedules after the section, exactly as the kernel's `thread::resume()` does, so the woken thread runs at once (the way PendSV does on ARM).
- **Build: the native** `platform-library.cmake` named a stale alias (`micro-os-plus::iii-posix-arch` → `micro-os-plus::posix-arch`); it linked `libucontext` (whose sources no longer compile against the current glibc — the port now prefers glibc's own `ucontext`), now dropped; and the kernel's `timegm.c` prototype had to be guarded (`__GLIBC__` / `__APPLE__` declare `timegm`, newlib does not) or the strict native flags turned the redundant redeclaration into an error. `test-cmsis-os-validator-host` added to the native config.
- **The last batch of platforms — now green.** `qemu-cortex-m0` / `m3` failed to build on `include/cmsis-plus/posix/dirent.h`: an empty `DIR` struct written as `{ ; }`, which `-Wextra-semi-Werror` rejects ("extra ';' inside a struct"). The stray `;` became a named `int reserved;`. Both then build and pass **3/3** (`rtos-apis`, `mutex-stress`, `cmsis-os-validator`), as do `qemu-cortex-m4f` / `m7f`. `nucleo-f767zi`, `nucleo-h743zi` and `raspberrypi-pico` build all three suites (3 `hwd` each); like `nucleo-f411re` their tests drive the board through OpenOCD, so they are now labelled `hwd` (a directory-level `set_tests_properties`) and the generic `test` action (`ctest -LE hwd`) skips them — `test-<platform>-cmake` runs them on a bench.

hwd tests are registered but never run — they need the board, one test per power cycle. They carry `LABELS hwd`, so `ctest -LE hwd` runs only the emulated/host set.

Per-platform notes (the recipe, applied)

- `platforms/{qemu-cortex-m0,m3,m4f,m7f,nucleo-f411re,nucleo-f767zi,nucleo-h743zi,raspberrypi-pico}` — the **Cortex-M** case, now **fixed**. Those platforms do not test a sibling port: they pull the *published* `@micro-os-plus/micro-os-plus-iii-cortexm` xpack (`v1.1.0`) plus the harness's own `device-qemu-cortexm`. The link failed with `undefined reference to Reset_Handler / NMI_Handler / ...` because the kernel's startup and exception handlers are an **optional group** (`micro-os-plus::iii-startup`), not part of the core, and the platform did not link it — nor the semihosting/newlib/posix-io groups the startup needs. Added to every Cortex-M `platform-library.cmake`:

```
cmake micro-os-plus::iii-startup # Reset_Handler, core handlers, startup.cpp micro-os-plus::iii-semihosting # os_startup_initialize_args, os_terminate micro-os-plus::iii-newlib-reent # _close, ... micro-os-plus::iii-posix-io # rtos-apis reaches POSIX-io
```

Verified: `qemu-cortex-m7f` and `qemu-cortex-m4f` build all three suites (`rtos-apis`, `mutex-stress`, `cmsis-os-validator`) and `mutex-stress-test` passes under QEMU.

The **loral** cortexm port's own boards are wired the same way, as `cortexm-<board>` platforms that reuse the port's builder. `cortexm-pico2` is done: **14 CTest cases** (2 `-qemu` + 12 `-hwd`) and it builds clean.

The QEMU split is **configuration**, done in the port, not in the tests: * `test/CMakeLists.txt` gained a per-test `BOARD_TEST_HWD_ONLY` list (a test there gets no `qemu` image even when the board has a QEMU linker); * `test/boards/pico2/board.cmake` gained `UOS_BOARD_LINKER_QEMU` (`linker-qemu.ld`, the generic Cortex-M33 memory map); * `test/pico2/tests.cmake` lists the USB pair and the multicore/SIO tests `HWD_ONLY`, leaving `exc-test`, `smp-test1`, `sc-test-ko` with a `qemu` image.

Caveat, and where it stopped: the RP2350's own blocks (SIO, USB, PSRAM) are not modelled by QEMU's generic Cortex-M33 machine, and making the three `qemu` tests actually run needs more than configuration. Work done toward it:

- a QEMU console glue, `test/boards/pico2/qemu/{include/bsp,src}/{uart,led}.hpp` (same `bsp/uart.hpp` API, implemented over the kernel's semihosting trace);
- per-variant hooks in the port's `test/CMakeLists.txt`: `board_test_qemu_libs/_sources/_includes` let the `qemu` image replace the board entirely;
- the pico2 `tests.cmake` uses them to link the kernel + the harness's `device-qemu-cortexm` (its `vectors-cortexm.c` owns the vector table) instead of the board;
- the `cortexm-pico2` platform adds `device-qemu-cortexm` and the `arm-cmsis` xpack, and the config inherits `arm-cmsis-dependencies`.

The remaining blocker: the port's **core** (`src/rtos/os-core.cpp` / `os-core-rp2350.cpp`) is RP2350-specific and is what provides `cmsis-plus/rtos/port/os-decls.h`; the `qemu` image links the kernel, not the board, so that header is missing (`fatal error: cmsis-plus/rtos/port/os-decls.h: No such file`). Running the emulated pico2 tests therefore needs a **generic Cortex-M33 port core** (a port refactor), not a configuration change.

Progress on that refactor (all done, build is green): * a generic port library `micro-os-plus::cortexm-qemu` in the port (the generic core `src/rtos/os-core.cpp` + the generic `include/`), with Cortex-M7 flags; * the emulated image targets QEMU's generic **Cortex-M7** machine (mps2-an500) -- the harness device's CMSIS core has no Cortex-M33; * `exc-test` (a bare-metal probe that defines its own `PendSV_Handler` / `SysTick_Handler`) is `HWD_ONLY`: it clashes with the device; * the emulated image takes the **device's** linker script, not the board's (`_variant_linker` is cleared when the board is replaced).

State now: `cortexm-pico2` **builds and runs green** with 2 `qemu` tests (`sc-test-ko-qemu`, `smp-test1-qemu`) + 12 `hwd`.

The runtime "hang" had two causes, both fixed: * **no output** — the kernel's `src/diag/trace-semihosting.cpp` only compiles under `OS_USE_TRACE_SEMIHOSTING_DEBUG` / `_STDOUT`; without it the weak no-op `trace::write` won and nothing was printed. The generic `micro-os-plus::cortexm-qemu` library now defines `OS_USE_TRACE_SEMIHOSTING_STDOUT` on its interface. * **never exiting** — the pico2 tests print `RESULT:` and then loop forever (a hardware convention), so a plain `qemu` CTest case always timed out. Added the port's `test/pico2/include/hw_result.hpp`: under `QEMU_BUILD` the test ends through the strong semihosting `_Exit()` (`src/semihosting-exit.cpp`) and CTest reads the exit code; on hardware it is a no-op (the pico2 `hw.sh` does not enable semihosting, so a hardware `_Exit()` would

halt on an unhandled BKPT) and the LED/heartbeat loop runs. `sc-test-ko` calls it after its RESULT line; `smp-test1` gained a bounded ten-beat self-check + RESULT before it.

Both now pass in a few seconds: `cortexm-pico2-sc-test-ko-qemu ... Passed 0.70 sec`, `cortexm-pico2-smp-test1-qemu ... Passed 3.24 sec`.

Hardware tests now end with a verdict too. The same `hw_result` helper applies to the `hwd` builds: every test that prints a RESULT line calls `hw_result::ok()/fail()` after it, so the run stops instead of looping. * `test/pico2/include/hw_result.hpp` now always ends through the strong semihosting `_Exit()` (both `qemu` and `hwd`; `SEMIHOST` is defined for every build of this port). * `test/boards/pico2/hw.sh` and `test/boards/pico2-pizero/hw.sh` enable semihosting on both cores before the resume and keep OpenOCD attached, reading the verdict from the log (`RESULT: PASS|FAIL`) instead of shutting down after the resume. Without semihosting the exit BKPT is an unhandled debug event and the core halts. * `smp-mat-test`, `smp-test4`, `smp-test5` (after a TX-ring drain) and `smp-test-ko` gained the call; `sc-test-ko` and `smp-test1` already had it. The probe tests without a verdict (`exc-test`, `smp-test0`, `smp-test2`, `smp-test3`, the USB pair) keep their idle loops.

cortexm-pico2-pizero — done (hardware only). The Pi-Zero RP2350B is the pico2 board plus pins/flash/probe and one real difference: it is built `__ARM_ARCH_8M_MAIN__`, and QEMU's generic Cortex-M machine has no Cortex-M33. So the board's `board.cmake` clears `UOS_BOARD_LINKER_QEMU` and the port's builder emits `hwd` alone; the platform registers 10 `-hwd` cases. Builds green; `xpm run test` finds nothing (no emulated set), as intended.

cortexm-pico2-rp2350b-psram — done. The WeAct RP2350B is the pico2 board plus pins/flash/PSRAM, on the same silicon (ARMv7E-M), so it keeps a `qemu` suite for the two emulatable tests (`sc-test-ko`, `smp-test1`) and lists the rest HWD_ONLY: the multicore tests need the SIO block QEMU does not model, and the PSRAM / nested-interrupt / clock-tree tests need the board's own memory map and peripherals. 16 cases (2 `qemu` + 14 `hwd`); builds green, both emulated tests pass.

The board test directories are independent copies, so the `hw_result` change was applied to the Pi-Zero and PSRAM copies too, and `hw_result.hpp` moved from `test/pico2/include/` to `test/boards/shared/` -- the one include directory every board's tests now get (test/CMakeLists.txt), so a new board does not need its own copy.

cortexm-nucleof411, **cortexm-weactf411**, **cortexm-weactf412 — done (hardware only).** The three STM32F4 boards are programmed over their ST-Link and run on the desk, so each board declares no `UOS_BOARD_LINKER_QEMU` and the port's builder emits `hwd` alone. The platforms register what it produced: * `cortexm-nucleof411` — 1 case (`mos-test1`). * `cortexm-weactf411` — 2 cases (`mos-test1`, `spi-pipeline`). * `cortexm-weactf412` — 2 cases (`mos-test1`, `uart-test1`). All build green. The `hw_result` verdict was added to their tests too: `spi-pipeline` prints its own PASS/FAIL and now ends with `RESULT: / hw_result`; `mos-test1` (all three) and `uart-test1` gained a bounded five-second run, a RESULT line and the semihosting exit. Their `hw.sh` already enabled semihosting, and now stays attached and reads the verdict instead of leaving OpenOCD running.

The Cortex-M family is now complete: `cortexm-pico2` (2 qemu + 12 hwd; 5 + 15 since the three harness suites were added as board apps, see below), `cortexm-pico2-pizero` (10 hwd), `cortexm-pico2-rp2350b-psram` (2 qemu + 14 hwd), `cortexm-nucleof411` (1 hwd), `cortexm-weactf411` (2 hwd), `cortexm-weactf412` (2 hwd). * the harness suites, now all wired: `rtos-apis` (needs `xpacks::chan-`

fatfs + micro-os-plus::iii-posix-io + its own os_main), mutex-stress, cmsis-os-validator (native 60/60, Cortex-M QEMU 3/3, hardware on nucleo-*), instrumentation and blinky (hardware, nucleo-f411re); and the AArch64 platform-support _Exit() fix above.

VS Code plugin — done. The plugin lists xpm actions, not CTest tests. Every configuration has a generic test that runs the whole emulated/host set (ctest -V -LE hwd), so the hardware cases are never run by it. The emulated/host platforms additionally have one action per test, test-<app>-<variant> (e.g. test-smp_test0-qemu, test-mutex-stress-host, test-cmsis-os-validator-host): native, aarch32-rpi-zero-2w, aarch32-luckfox-lyra, aarch64-rpi-zero-2w, and the six cortexm-* platforms. The upstream hardware platforms (qemu-cortex-m*, nucleo-*, raspberrypi-pico) have only the generic test plus a top-level test-<platform>-cmake that runs all their cases with a bare ctest -V (needed because their tests are all hwd, which the generic test skips). Each action configures with the pinned toolchain ({ properties.commandCMakePrepareWithToolchain }), not the bare commandCMakeReconfigure so a fresh tree does not pick the host compiler.

Verified: xpm run test-smp_test0-host --config native-cmake-gcc-debug → native-smp_test0-host ... Passed; and xpm run test-sc-test-ko-qemu --config cortexm-pico2-cmake-gcc-debug → cortexm-pico2-sc-test-ko-qemu ... Passed.

Cortex-M33 QEMU platforms (pico2-1cpu & 2xcortex-m33) & SMP cmsis-os-validator — 100% green. * pico2-1cpu (QEMU mps2-an505, 1x Cortex-M33): 3/3 tests pass (rtos-apis, mutex-stress, cmsis-os-validator 60/60). * 2xcortex-m33 (QEMU mps2-an521, 2x Cortex-M33 SMP): 3/3 tests pass (rtos-apis, mutex-stress, cmsis-os-validator 60/60). * **SMP fixes for cmsis-os-validator:** 1. **Running thread re-enqueue guard:** thread::resume() only enqueues threads whose state is state::suspended or state::initializing. On SMP, a running thread has ready_node_.next() == nullptr; previously, raising signals/flags (flags_raise()) invoked resume() which linked the actively executing thread back into ready_threads_list_, allowing the secondary core to pick and execute the same thread concurrently on the identical stack. 2. **Thread attributes CPU affinity:** Added th_cpu_affinity to thread::attributes and os_thread_attr_t under OS_USE_SMP_SCHEDULER, passed into thread::thread constructor to ensure threads are created with affinity before internal_construct_ / resume() places them on the ready list. 3. **CMSIS-RTOS single-core compatibility:** CMSIS-RTOS v1 is fundamentally single-core; osThreadCreate and os_main_thread pin threads to Core 0 (1u << 0). This ensures strict priority preemption (TC_ThreadPriorityExec and TC_MutexPriorityInversion) and guarantees that per-core private NVIC registers on Cortex-M handle test interrupts on the core that enabled them (TC_ThreadInterrupts). 4. **SMP stack pointer invariant:** In switch_stacks(sp) (os-core-m33.cpp and os-core-rp2350.cpp), when a core continues executing the same thread (new_thread == old_thread), old_thread->context_.port_.stack_ptr = nullptr is cleared so secondary cores do not consider the live context switchable.

The harness suites as cortexm-pico2 board apps — done. rtos-apis, mutex-stress and cmsis-os-validator also run as pico2 applications (test/pico2/<suite>/, a placeholder harness-suite.cpp naming each, and board_test_libs() in test/pico2/tests.cmake bringing test::<suite>). That had been committed half-done; four things finished it:

1. **The emulated image is single-core.** The port's builder (test/CMakeLists.txt) built a -qemu image at the board's NCPU (2), but when board_test_qemu_libs replaces the board the core is the

generic single-core `micro-os-plus::cortexm-qemu`, and a 2-CPU build does not compile against it (`os-core.cpp:497/833`, `current_thread_`). Such an image is now NCPU=1. It also kept only the board replacement on its link line and dropped the suite's library; it now keeps it.

2. **One interface for what a harness platform would give.** `pico2-harness-suite` in `tests.cmake`: `OS_USE_OS_APP_CONFIG_H` (so the suite's config is read), the harness platform's include (its `cmsis-plus/platform.h`), the POSIX pair the semihosting syscalls need, the kernel's semihosting / newlib / posix-io groups (the `hwd` image links only the SoC), and `-Wl,--wrap=os_main`.
3. **The verdict.** A suite returns its code from `os_main()`; `hw.sh` reads a `RESULT` line. Each placeholder now defines `__wrap_os_main()`: run the suite, print `RESULT: PASS|FAIL` as the other pico2 tests do, return the code to the kernel's `std::exit()` -- the same `SYS_EXIT` `hw_result` ends with.
4. **The suite's configuration wins.** The board's `os-app-config.h` is first on the `hwd` include path; under `UOS_HARNESS_SUITE` it now includes the suite's first and fills only what the suite leaves unset. A board test defines no `UOS_HARNESS_SUITE` and gets exactly the old values.

Verified: `cortexm-pico2` builds 20/20; `xpm run test --config cortexm-pico2-cmake-gcc-debug` passes **5/5** (`cmsis-os-validator-qemu` 60/60, `mutex-stress-qemu`, `rtos-apis-qemu`, `sc-test-ko-qemu`, `smp-test1-qemu`). No existing image moved: the 14 earlier pico2 images and all 31 of the other five `cortexm-*` boards are byte-identical to the previous commit, and the `rp2350b-psram` emulated pair still passes. The three `-hwd` suite images build but have not been run on a board.

Multi-Architecture & SMP/1-CPU Test Verification Matrix (100% Green)

All three base test suites (`mutex-stress`, `rtos-apis`, `cmsis-os-validator`) have been verified across QEMU SMP and 1 CPU architectures, as well as POSIX host:

Platform / Board	Machine / CPU	Mode	<code>mutex-stress</code>	<code>rtos-apis</code>	<code>cmsis-os-validator</code>	Status
<code>pico2-1cpu</code>	QEMU MPS2 AN505 (Cortex-M33)	1 CPU	PASSED (28.3s)	PASSED (4.7s)	PASSED (60/60)	PASS
<code>2xcortex-m33</code>	QEMU MPS2 AN521 (Cortex-M33)	2 CPU SMP	PASSED (28.4s)	PASSED (4.7s)	PASSED (60/60)	PASS
<code>native</code>	POSIX synthetic SMP (Host)	1 CPU / Host	PASSED (15.2s)	PASSED (7.2s)	PASSED (60/60)	PASS
<code>qemu-cortex-m0</code>	QEMU MPS2 AN385 (Cortex-M0)	1 CPU	PASSED (23.7s)	PASSED (4.7s)	PASSED (2.6s)	PASS
<code>qemu-cortex-m3</code>	QEMU MPS2 AN385 (Cortex-M3)	1 CPU	PASSED (23.7s)	PASSED (4.7s)	PASSED (2.3s)	PASS
<code>qemu-cortex-m4f</code>	QEMU MPS2 AN386 (Cortex-M4F)	1 CPU	PASSED (22.9s)	PASSED (4.7s)	PASSED (2.3s)	PASS
<code>qemu-cortex-m7f</code>	QEMU MPS2 AN500 (Cortex-M7F)	1 CPU	PASSED (22.8s)	PASSED (4.7s)	PASSED (2.4s)	PASS
<code>aarch32-rpi-zero-2w</code>	QEMU raspi3b (Cortex-A53)	4 CPU SMP	PASSED (36.0s)	PASSED (7.4s)	N/A (NVIC-only)	PASS
<code>aarch64-rpi-zero-2w</code>	QEMU raspi3b (Cortex-A53)	4 CPU SMP	PASSED (35.9s)	PASSED (7.3s)	N/A (NVIC-only)	PASS

Note on Architecture Support: * `cmsis-os-validator` is specific to Cortex-M NVIC (uses `NVIC_EnableIRQ`, `IRQn_Type`) and POSIX (which provides simulated signal-based NVIC shims). It is not applicable to bare-metal AArch32/AArch64 GIC architectures. * In QEMU, `raspi3b` requires a minimum of 4 CPUs (`-smp 4`) per the BCM2837 SoC definition. * `device-qemu-cortexm/include/cmsis-plus/cortexm/exception-handlers.h` had multi-line comment warnings (`-Werror=comment`) caused by trailing backslashes on single-line comments, which was fixed and committed.

Commands that work today

```
cd "$HOME/TMP/micro-os-plus-iii-smp.git/tests"

# AArch32 Pi — the whole emulated set (port tests + mutex-stress + rtos-apis)
xpm run test --config aarch32-rpi-zero-2w-cmake-gcc-debug

# one suite only
xpm run test-rtos-apis-test-qemu --config aarch32-rpi-zero-2w-cmake-gcc-debug

# POSIX-arch on the host
xpm run test --config native-cmake-gcc-debug
```

`xpm run test` sets the toolchain/QEMU `PATH` and passes `-LE hwd`, so the hardware cases are never run. A bare `ctest -LE hwd` in the build folder does the same filtering but needs `build/<config>/xpacks/.bin` on `PATH` for the emulator.

What a fresh clone gives you

The tests are **source**, and the sources are kept — but only if the commit includes the **untracked** files. A plain `git commit -a` would not: it commits modified *tracked* files only, and the new platforms are untracked. So commit with `git add -A`.

Untracked today (absent from a clone unless added):

- `micro-os-plus-iii-smp.git/tests/platforms/cortexm-{pico2,pico2-pizero,pico2-rp2350b-psram,nucleof411,weactf411,weactf412}/` — the six new platforms, five files each;
- `micro-os-plus-iii-cortexm.git/test/boards/pico2/linker-qemu.ld`, `.../test/boards/pico2/qemu/{include/bsp,src}/*` (six files), and `.../test/boards/shared/hw_result.hpp`.

Already tracked, so committed with the rest: the shared runner `micro-os-plus-iii-smp.git/test_smpl/` (`run-qemu.sh`, `run-hw.sh`, `run-host.sh`, `hw_result.hpp`, `board-contract.cpp`), the harness device `tests/device-qemu-cortexm/`, the ports' own test sources, and every `tests/platforms/*` CMakeLists.

Never in the repo, by design:

- `build/<config>/` — gitignored (`build*/`); `xpm run install` / `prepare` / `build` regenerate it;
- `tests/xpacks/` and `node_modules/` — installed by `xpm install` / `npm install`; the QEMU, CMake, Ninja, build-helper, chan-fatfs, validator and SEGGER packages are symlinks into `~/.local/xPacks`;
- hardware: an `hwd` test is registered but needs the real board and OpenOCD.

So, from a clone to green:

```

# 1. the six repositories, beside each other, as `*.git` (the ports find them
#    by either name, see Step 2)
cd "$HOME/TMP"
for r in aarch32 aarch64 cortexm devices posix-arch smp; do
    git clone "$HOME/Downloads/GIT/micro-os-plus-iii-$r.git" \
        "$HOME/TMP/micro-os-plus-iii-$r.git"
done

# 2. the tools (xpacks, including QEMU)
cd micro-os-plus-iii-smp.git/tests
npm install
xpm install

# 3. one configuration, end to end -- install, prepare, build, test, in that
#    order; `test` alone does not build (see Step 4)
xpm run install --config aarch32-rpi-zero-2w-cmake-gcc-debug
xpm run prepare --config aarch32-rpi-zero-2w-cmake-gcc-debug
xpm run build   --config aarch32-rpi-zero-2w-cmake-gcc-debug
xpm run test    --config aarch32-rpi-zero-2w-cmake-gcc-debug # emulated set

# or a whole emulated family, four steps each
for c in qemu-cortex-m0 qemu-cortex-m3 qemu-cortex-m4f qemu-cortex-m7f; do
    C="$c-cmake-gcc-debug"
    xpm run install --config "$C" && xpm run prepare --config "$C" && \
    xpm run build   --config "$C" && xpm run test    --config "$C"
done

# the same set on release -- a separate build folder, so its own install too
for c in qemu-cortex-m0 qemu-cortex-m3 qemu-cortex-m4f qemu-cortex-m7f; do
    C="$c-cmake-gcc-release"
    xpm run install --config "$C" && xpm run prepare --config "$C" && \
    xpm run build   --config "$C" && xpm run test    --config "$C"
done

```

The `qemu` cases then run under QEMU; the `hwd` cases are listed but skipped (`-LE hwd`) until a board is attached.